Nepal Telecom Churn Analytics Executive Report

AI-Driven Churn Explanations & Segment Analysis

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Scope: Filtered Customer Segment (50 Subscribers)

1.

Key Performance Indicators (KPIs):

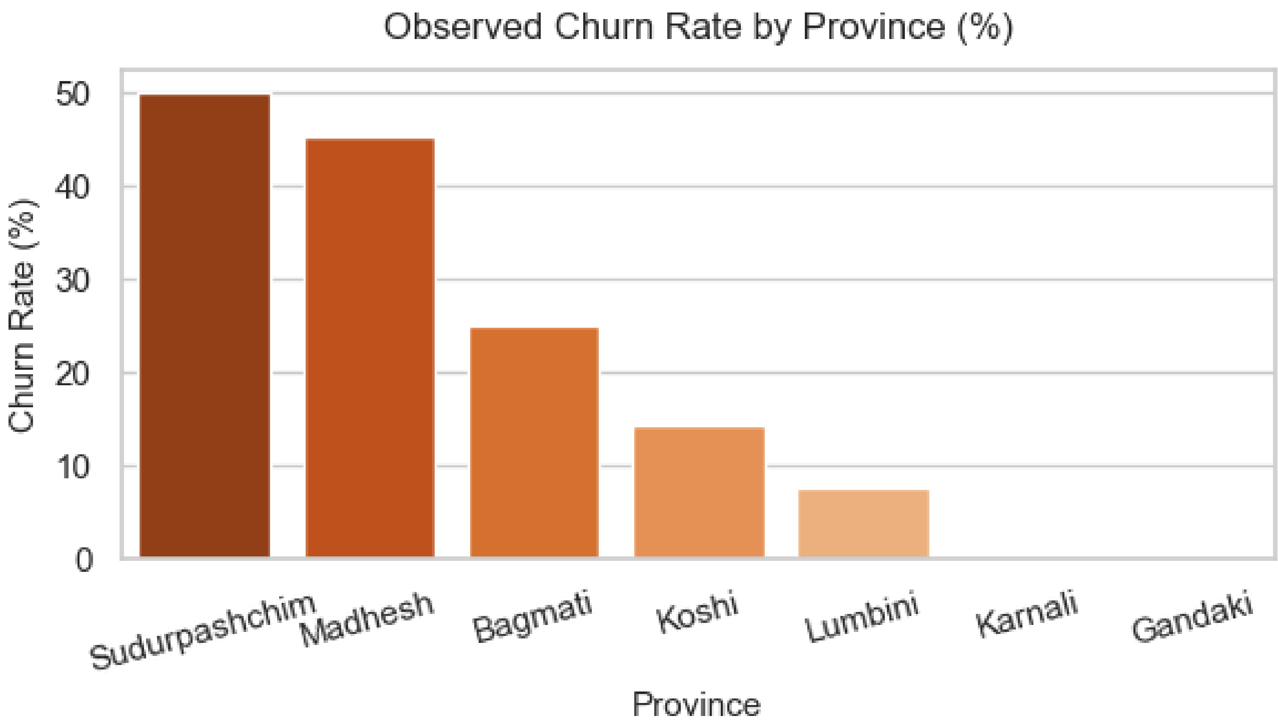
* Total Evaluated Customers: 50
* Segment Churn Rate (Observed Churn): 22.00%
* Model Prediction Accuracy (XGB): 85.0%
* Monthly Revenue At Churn Risk: Rs. 1,820.00

Note: Model Strictness set to 'Medium Strictness' (Churn Cutoff: 50%).

How to Use This Report

Use the charts and interpretations in this document to identify structural vulnerabilities in service quality, geographical regions requiring network reinforcement, and customer behavior patterns indicating imminent churn. Please note that the comprehensive list of at-risk customers has been exported to the companion Excel spreadsheet for direct telecommunication campaigns.

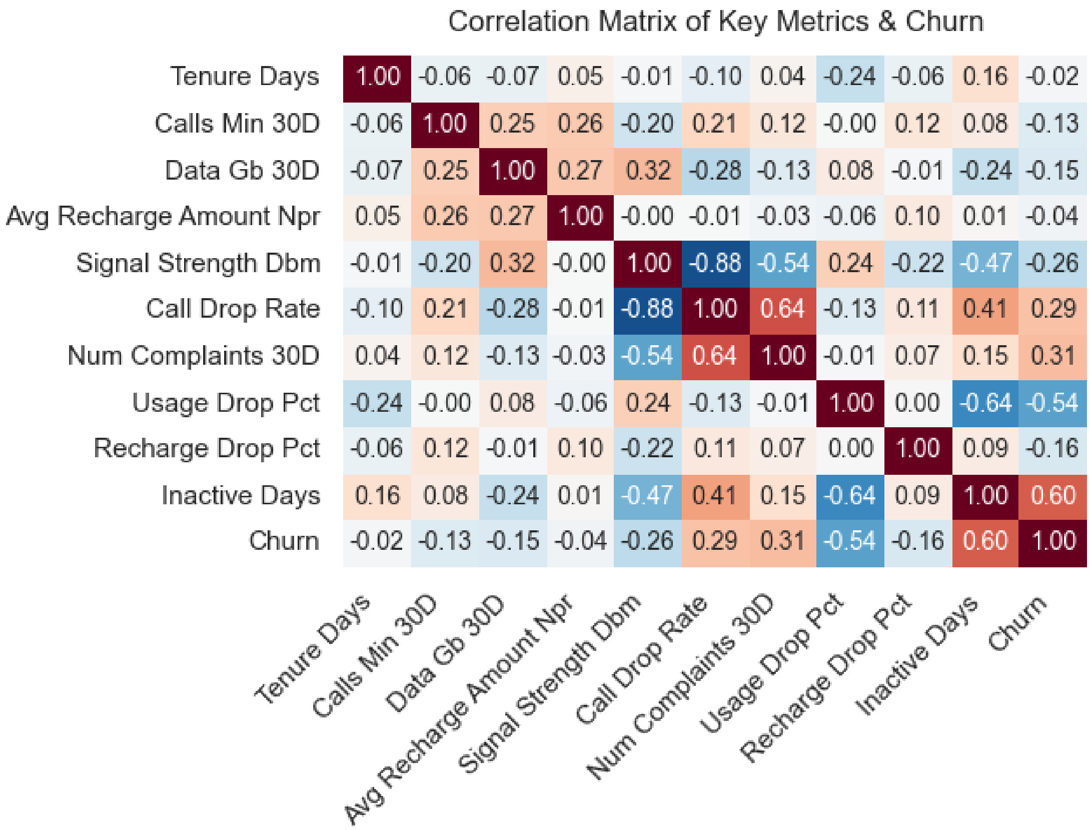
2. Regional Churn Analysis



Interpretation & Business Actions:

This chart represents the observed rate of customer churn across the different provinces of Nepal. By understanding how churn is geographically distributed, regional managers can identify provinces that may be suffering from network infrastructure issues, low coverage, or localized competitive threats. High churn rates in specific provinces highlight a critical need for targeted network expansion, regional promotions, and local dealer engagement to retain subscribers.

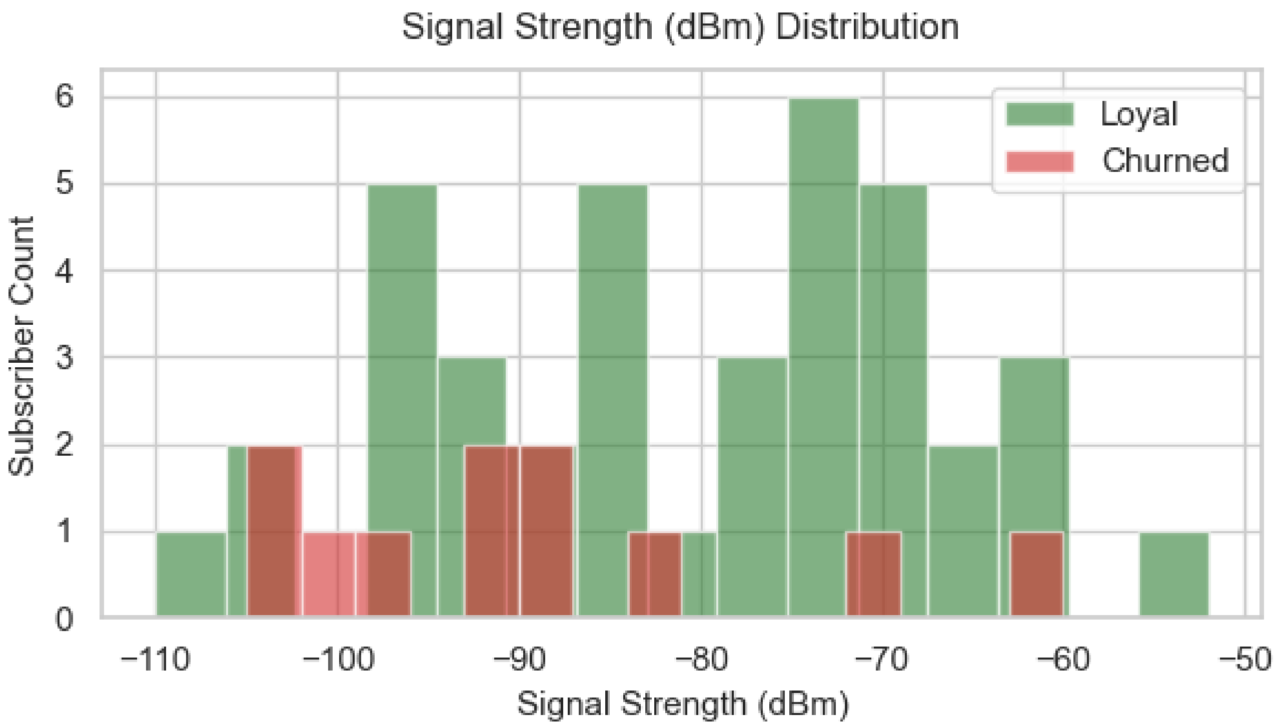
3. Linear Feature Correlations



Interpretation & Business Actions:

The correlation matrix calculates the linear relationship between key customer metrics (such as tenure, call drops, complaints, and recharge behavior) and the final churn outcome. The values range from -1.0 to +1.0. A positive value (red) indicates that as the feature increases, churn increases (e.g., number of complaints or inactive days). A negative value (blue) indicates that as the feature increases, churn decreases (e.g., higher average recharge amount or longer tenure). This matrix is essential for identifying the strongest predictors of churn at a glance.

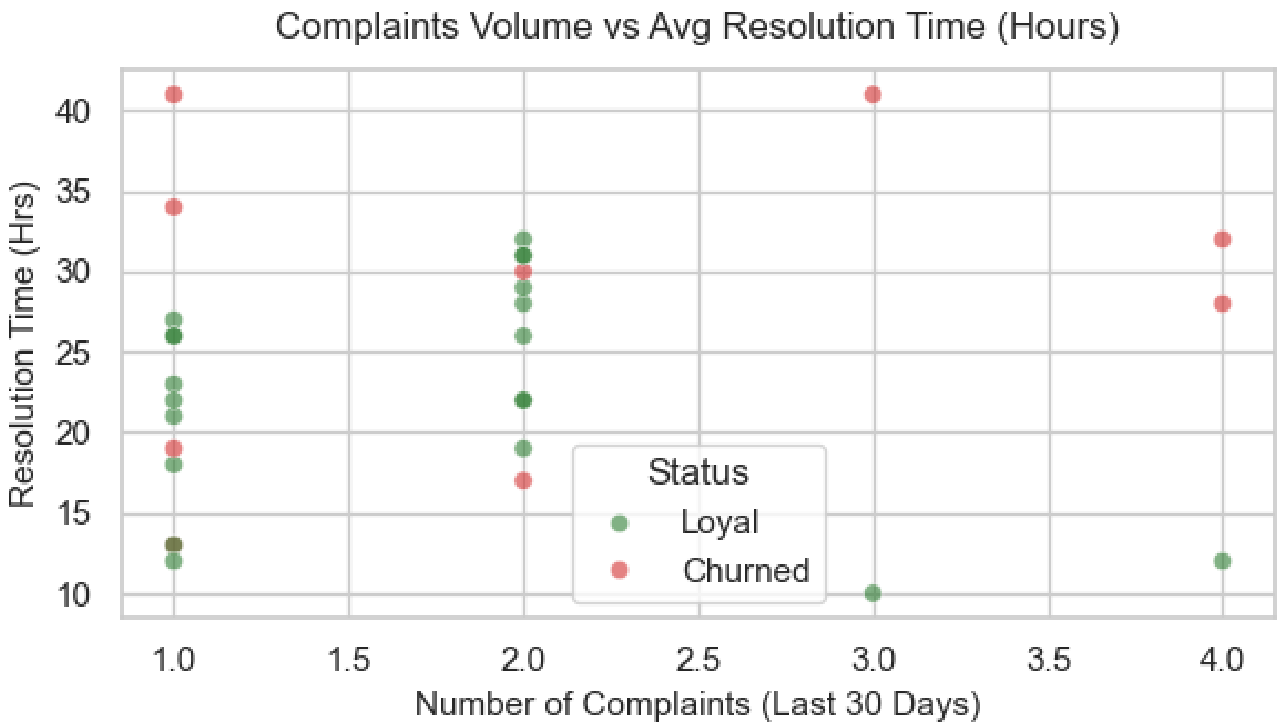
4. Network Quality & Signal Strength Distribution



Interpretation & Business Actions:

This histogram shows the distribution of signal strength (measured in dBm) for both loyal (green) and churned (red) customers. Network signal strength typically ranges from -50 dBm (excellent) to -110 dBm or lower (very poor/no signal). A clear shift of churned customers towards the lower end of the signal strength spectrum indicates that poor network quality is a primary driver of customer dissatisfaction and churn. Addressing coverage blindspots in these areas can directly improve customer retention.

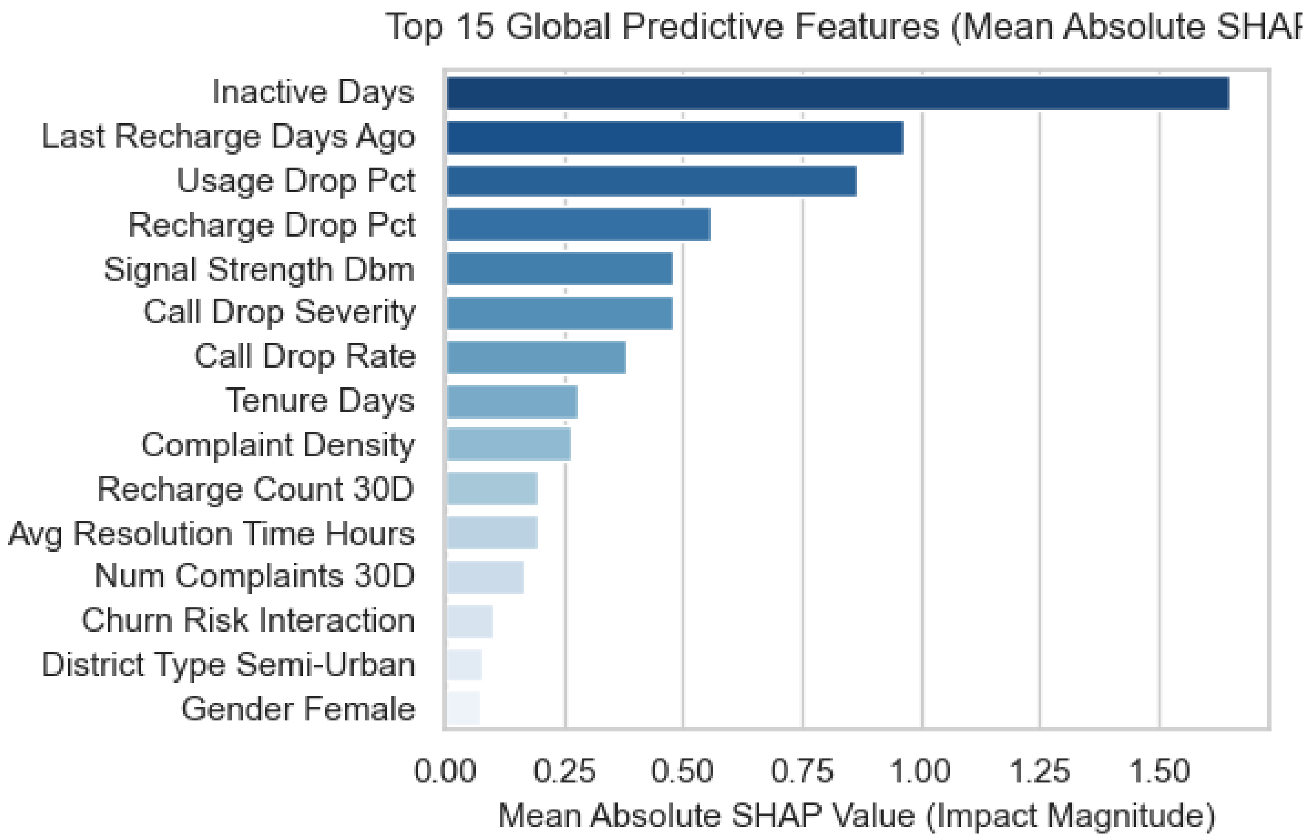
5. Service Quality & Complaint Resolution



Interpretation & Business Actions:

This scatter plot visualizes customer complaints against the average resolution time in hours, colored by churn status. Points clustered in the upper-right quadrant represent customers with many complaints that took a long time to resolve. A high concentration of churned customers in this area highlights the direct negative impact of delayed service resolution on customer loyalty. Streamlining the ticketing system and reducing resolution times for high-value customers is a crucial retention strategy.

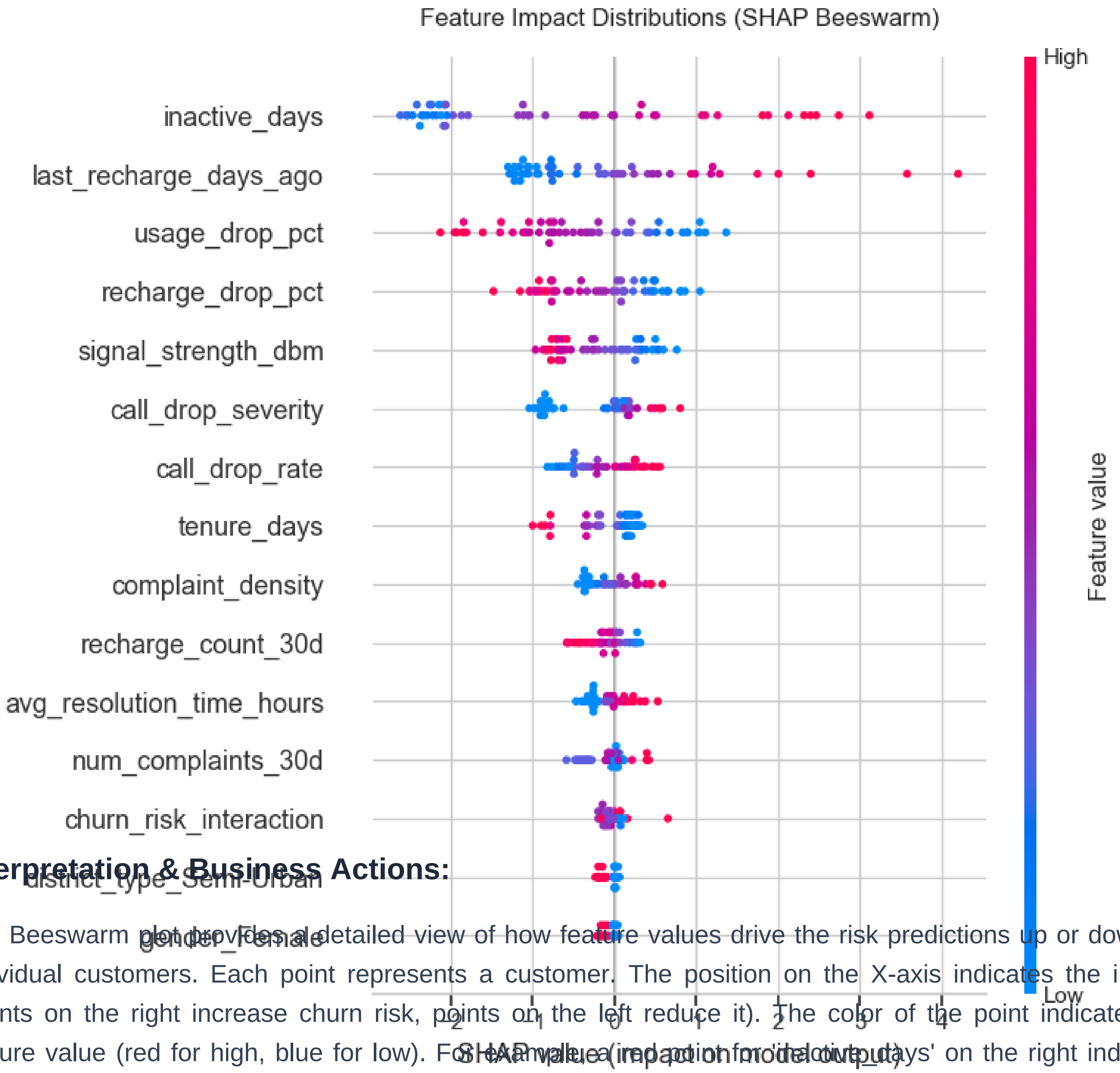
6. AI-Driven Global Feature Importance



Interpretation & Business Actions:

This chart showcases the top 15 features that have the highest overall influence on the machine learning model's churn predictions, ranked by their Mean Absolute SHAP value. SHAP (SHapley Additive exPlanations) values measure the direct contribution of each feature to the model's output. Features at the top of the chart represent the most critical indicators that the AI looks at when determining whether a customer is likely to churn. Business units should base their predictive triggers on these high-ranking attributes.

7. Feature Distributions & Risk Impact (Beeswarm)



Interpretation & Business Actions:

The Beeswarm plot provides a detailed view of how feature values drive the risk predictions up or down for individual customers. Each point represents a customer. The position on the X-axis indicates the impact (points on the right increase churn risk, points on the left reduce it). The color of the point indicates the feature value (red for high, blue for low). For example, a red point for 'inactive_days' on the right indicates that high inactivity heavily drives up churn risk, whereas a blue point indicates low inactivity reduces risk.

8. Risk Dependency Analysis

Interpretation & Business Actions:

The feature dependence plot illustrates how the risk impact (SHAP value) of a single feature (in this case, customer age) varies across its range of values. Points below the zero line indicate age groups that are more likely to remain loyal to the network, while points above the zero line indicate age groups with a higher risk of churn. This allows marketing teams to design age-specific retention packages, such as youth-centric data bundles or senior-citizen loyalty programs, based on actual historical behavior.